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Project Title:	The Impact of Real Microgravity on the Proliferation of Human Neural Stem Cells and derived-Oligodendrocytes		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
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Space Biology Element:	(1) Cell & Molecular Biology		
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Task Description:	Recent data from NASA shows that astronauts present vision, physiological, and anatomical changes as well as intracranial pressure (ICP) and visual impairment intracranial pressure (VIIP) (more recently known as Spaceflight Associated Neuro-ocular Syndrome (SANS)) after long-duration space missions. These changes are unlike any clinical entity previously seen. Yet, the cause(s) of chronically elevated ICP/VIIP and potential consequences are unknown. The main goal of this International Space Station (ISS) spaceflight proposal is to obtain reference data on the effects of real microgravity (0G) to validate and expand our recent data obtained using simulated microgravity (sim-μG). We want to assess the impact of real microgravity on the cell cycle of human neural stem cells and derived oligodendrocytes (OL) that are the myelin forming cells in the central nervous system. We now have evidence that sim-μG increases the number of neural progenitors and OL progenitor (OLP) via cell proliferation. Here we postulate that microgravity-associated VIIP and altered ophthalmic anatomy (papilledema, choroidal folds, etc.) as well as a potential microgravity-associated ICP without visual impairment might be at least in part due to increased cell proliferation in the brain. Aim I. To test the hypothesis that neural stem cells (NSC) and OL progenitors exposed to real microgravity will proliferate faster and will give rise to more progenies than cells grown in 1G. Aim II. To test the hypothesis that in real microgravity NSC and OL produce and secrete one or more molecules that may act as regulators of cell proliferation. Significance. Neural cells possess intrinsic plasticity that can be modulated by the environment and their response to real microgravity may confirm and even reveal new properties of these cells triggered by it. We also propose to examine molecules secreted by these cells while in space in order to identify potentially unique regulators of cell proliferation. It is known that secretome proteins play a key role in cell signaling and cell migration. These molecules may also hold a potential key for the modulation of normal cell proliferation for future therapeutic interventions for the central nervous system (CNS). Approach. Human NSC and OLP will be plated using our proprietary specific culture media in self-contained hardware designed for 0G (Astrium). Control cells will be maintained in 1G. Ground based experiments. We will analyze the effects of real microgravity on NSC and OL cell cycle via immunocytochemistry. After landing and recovery of the samples, the conditioned culture medium will be collected, maintained at 4°C, and transported to our laboratory to further characterize the secretome of both cell types maintained in real microgravity. Our stem cell culture medium as well as the oligodendrocyte specification medium are both serum-free and contain few proteins, properties that will facilitate the secretome analysis. The characterization of the secretome will include identification of proteins and small molecules that may be produced and secreted in real microgravity only. Innovation. This is the first study on the impact of real microgravity on human neural cells. Our data will provide valuable information to better understand the gravitational epigenetic modulation of human neural cells. Benefits for Humanity. Confirming that real microgravity increases cell proliferation will move us forward toward a better understanding on the causes of ICP and VIIP in astronauts. Moreover, since long-term missions in space are imminent, the identification of molecules that increase cell proliferation might lead to the design of potential countermeasures to prevent or alleviate the negative effects observed in astronauts exposed to long-term missions in space. Myelin is responsible for the rapid conduction of electrochemical messages to and from the brain. Since there is no cure for myelin disorders our data might also lead to novel therapeutic modalities for myelin repair.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	To date, there is no cure for developmental disabilities, neurodegenerative disorders, myelin loss, or cancer. We expect that our work described in the current proposal will generate cures that are of paramount importance in two different but complementary fields: 1) Drug discovery: The first product will be the first molecule that would have been identified, isolated, and tested to be responsible for increased cell proliferation. This will open the door for a second product (or series of products) under this category that would be an antibody or similar molecule(s) that would void/neutralize the effects of the first molecule on cell proliferation. This second product will be further studied as an anti-cancer agent via prevention of cell proliferation. If there is more than one relevant secreted molecule a similar process would be followed. Once we have data showing that we have interesting molecules, we will submit an invention report to the Office of Intellectual Property at UCLA (University of California, Los Angeles). This office supports scientists by contacting potentially interested companies and helping disseminate information on technologies available for licensing. In some cases UCLA technologies are commercialized by large companies. Alternatively, some entrepreneurs-inventors form successful startups based on their own UCLA technology. 2) Generation of cells for cell replacement therapies: The presence of functional myelin is essential for all our normal daily activities. Our overall goal is to investigate the prospects of cell replacement therapies for patients that require normal OL and myelin to restore proper CNS function. We contemplate to be able to apply this approach to propagate OLPs also known as (OPCs) and make them available to treat the spectrum of myelin disorders, from white matter damage in the premature neonate and infants suffering of autism or maternal prenatal infection, to the patient with leukodystrophies, multiple sclerosis, psychosis/schizophrenia, and trauma. We have previously demonstrated that OLPs are cells that, once grafted, migrate, integrate, and interact with the host CNS. Validating the functionality of OLPs exposed to real μG will open a new era on the ways we conceive cell replacement therapies and CNS repair. Moreover, trauma to the brain and spinal cord involves myelin loss and glial and neuronal cell death; thus, data generated by the current study will provide insights on producing cells to be transplanted into the CNS of individuals suffering from trauma. Prior to commercializing the generation of cell populations in microgravity, it is essential to confirm that microgravity-exposed cells are fully functional and safe; by biochemical and pre-clinical neural transplantation proposed in this grant application we will generate the necessary evidence. Once this step will have been accomplished, we'll

	contact specific companies that might provide us with the revenue necessary to fund the scale-up process to generate human induced pluripotent stem cell (hiPS) derived oligodendrocytes progenitors (OLP). By then we hope to have created strong partnership with the industry in order to be able to generate cells to overcome neural degeneration and developmental disabilities.
Task Progress:	We examined by continuous live imaging the behavior of space flown neural stem cells (NSCs). We found two cell sub-populations in the cultures of space (SPC)-flown NSCs, distinguished by differences in cell diameter. We also found two subpopulations in ground control (GC) cells. Nonetheless, the effect SPC-µG resulted in a shift of both sub-populations. The small and large NSCs grown in Earth gravity were evenly distributed; large NSCs accounted for 49.2% and small NSCs represented 50.8% of the entire population. In contrast, the proportion of cells included in each group shifted after exposure to SPC-µG, where large cells represented 81% of the total number of cells while the percentage of small NSCs had decreased to 19%. Thus, our study shows that SPC-µG generated a greater proportion of 'large' NSCs than ground control cells. This study is significant to astronauts' health because it shows for the first time that SPC-µG increase the proportion of "large" NSCs after space flight. Moreover, space flight also increased the proliferation of NSCs when compared to ground control cells, testing our hypothesis that microgravity generates more cells than Earth gravity. Understanding the effects of microgravity on neural cells will facilitate the design of measures to ensure astronaut health and safety on longer space missions and whilst at the International Space Station (ISS) and beyond. To better understand these changes, we examined the molecules secreted by the cells while in space. A total of 119 differentially expressed proteins were identified by analysis of the secretome of NSCs, 86 upregulated in SPC-NSCs/GC-NSCs and 34 downregulated. From those, 37 upregulated proteins have a fold change = 2 and 5 downregulated proteins a fold change = -2. The changes involved proteins implicated in activation of endoplasmic reticulum (ER) stress response, Secreted Protein Acidic and Cysteine Rich (SPARC), Calreticulin (CALR), and Endoplasmin (HSP90B1), which were found to be 12.8, 5.2, and 3.3 times, respectively, more expressed in SPC-NSCs than GC-NSCs. We are in the process of analyzing how the protein changes co-relate with the physical changes observed in terms of cell size. From SpaceX CRS-21 (SpX-21), we published our first paper. Main Findings: We saw that oligodendrocytes, the cells that make and maintain myelin through the life of the organism, displayed a more immature morphology and ontogenic profile with respect to ground control cells. This phenomenon was previously observed in simulated microgravity. Since these studies were performed after space flight, they indicate that oligodendrocyte (OL) progenitors (OLPs) remember having been in space. While we do not know if these changes occur in the natural milieu of the entire brain where various cell populations co-exist, having immature OLs may lead to some myelin loss. This would lead to cognitive and mechanical changes due to myelin loss. More studies are necessary before one can reach any conclusions.
Bibliography Type:	Description: (Last Updated: 02/07/2024)
Articles in Peer-reviewed Journals	Biancotti JC, Carpo N, Zamudio J, Vergnes L, Espinosa-Jeffrey A. "Profiling the secretome of space traveler human neural stem cells." J Stem Cell Res Dev Ther. 2022 Jun 10;8:094. http://dx.doi.org/10.24966/SRDT-2060/100094 , Jun-2022
Articles in Peer-reviewed Journals	Tran V, Carpo N, Shaka S, Zamudio J, Choi S, Cepeda C, Espinosa-Jeffrey A. "Delayed maturation of oligodendrocyte progenitors by microgravity: Implications for multiple sclerosis and space flight." Life (Basel). 2022 May 27;12(6):797. https://doi.org/10.3390/life12060797 ; PMID: 35743828; PMCID: PMC9224676 , May-2022